

A Project Presentation on Chat App

Group no: 11

Enrollment Nos :24034211051

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INTRODUCTION

- **Chat App** is a mobile-based messaging application designed to enable fast, secure, and real-time communication between users.
- The application allows users to **send text messages, voice notes, images, videos, and documents, voice and video calls—all from a single, easy-to-use platform.**
- The system aims to replace **traditional SMS and fragmented communication tools with a modern, unified, and efficient digital communication solution.**

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1.1 Project Profile

Field	Details
Project Title	Chat App
Objectives	• To provide a digital platform for instant and real-time communication between users. • To enable secure text messaging, voice notes, image, video, and document sharing. • To provide secure login using mobile number and OTP verification.
FrontEnd	• Technology: React Native (Expo Framework)
BackEnd	• Technology: Node.js with Express.js • Database: MongoDB • Authentication: OTP-based mobile authentication
GroupNo	11
Enrollment No	24034211051
Company Name	Bytesved Technologies
Internal Guide	Ms. Priyansi Patel

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1.2 About the Organization

- **Bytesved Technologies LLP** is a dynamic and innovative software development company specializing in delivering modern digital solutions to businesses worldwide.
- The company is headquartered in **Gandhinagar, Gujarat, India** and operates with a vision to help organizations transform ideas into high-performance software products.
- Bytesved provides a comprehensive range of **IT and digital services**, including: **Mobile App Design & Development, Website Design & Development, Branding & Identity Design, Creative Consulting & Concept Development**

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SYSTEM STUDY AND ANALYSIS

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2.1 Problem Statement

Traditional Chat involves:

- Traditional communication methods like SMS and email are slow and limited.
- SMS has no multimedia support and high costs.
- Lack of real-time interaction in emails.
- No centralized chat history across devices.
- Security and privacy issues in basic communication systems.

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2.2 Existing System

- Existing systems are:
 - SMS (Short Message Service)
 - Phone calls
 - Email communication
 - Social media messaging
- Examples:
 - Traditional SMS texting
 - Manual phone calls for communication
 - Basic messaging without encryption or advanced features

2.2.1 Drawbacks

- High dependency on internet connectivity
- Limited or no end-to-end security in traditional systems
- Risk of data loss when messages are not backed up
- Time-consuming communication through calls or emails
- Difficult to manage group conversations efficiently
- No secure authentication in basic messaging methods

2.3 Proposed System

- ❖ The Proposed Chat App Application Provides:
 - User registration and login using mobile number with OTP authentication
 - Real-time one-to-one and group chat functionality
 - Secure messaging with end-to-end encryption
 - Online/offline status and message delivery indicators
 - Voice and video calling features
 - Centralized and cloud-based data storage

2.3.1 Advantages or Comparison Study

Existing System	Proposed System
SMS, calls, or emails	Real-time chat application
No centralized chat history	Centralized chat storage
Limited multimedia support	Supports images, videos, audio & documents
No or weak security	Secure authentication & encrypted chats
Time-consuming communication	Instant messaging

2.3.2 Modules (with short description)

- **Authentication Module**:- Login using mobile number & OTP
- **User Profile Module** – Manage user details such as name, profile photo, and status.
- **Contact Management Module** – Add, sync, and manage contacts for chatting.
- **One-to-One Chat Module** – Enables real-time private messaging between users.
- **Group Chat Module** – Create and manage group conversations with multiple users.
- **Voice & Video Call Module** – Supports real-time audio and video calling features.
- **Security Module** – Ensures encrypted communication and data protection.

Hardware Requirement

Component	Server	Client
Processor	2.5 GHZ	1.6 GHZ
RAM	8 GB	2 GB
Hard Disk	10 GB free space	5 GB free space
Other	Browser	Browser

Software Requirement

Component	Server	Client
OS	Windows NT/Linux	Windows 10
Database	MongoDB 6.0	-
Browser	Chrome 114/ Edge 100+	Chrome 114+

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About Front-End

- **Technology:** React.Native with Expo

Usage:-Build a cross-platform mobile UI
Smooth navigation and user Experience

- React Native Stylesheet

Usage:-For modern and customize styling a webpage

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About Back-End

- **Technology:** Node.js version 20.0 with Express.js version 4.18

Usage:-Scalable server-side application with handle routing and APIs.
Secure Authentication

- **MongoDB:** MongoDB version 6.0

Usage:-No SQL database for storing data in JSON format.
Store Weddings,guest and chandla data

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